

Square-zero local Banach algebras satisfying von Neumann's inequality are isometrically operator algebras

Partial-result packet for arXiv:1804.04062

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Status: candidate partial result, likely valid. The general question remains open. The result below gives a complete affirmative answer for every normalized unital algebra $\mathcal{A} = \mathbb{C}1 \oplus X$ whose closed codimension-one ideal satisfies $X^2 = 0$, without a dimension assumption.

1 Source question

Badea, Crouzeix, and Klaja ask whether every normalized unital complex Banach algebra satisfying the one-variable von Neumann inequality is isometrically isomorphic to a closed algebra of Hilbert-space operators [1, p. 4]. Explicitly, their hypothesis is

$$\|a\| \leq 1 \implies \|p(a)\| \leq \max_{|z| \leq 1} |p(z)| \quad (p \in \mathbb{C}[z]).$$

The source passage is reproduced below.

We start by introducing the definitions, notations and prior results that we shall need.

Banach algebras satisfying von Neumann inequality. A unital complex Banach algebra $\mathcal{A}$ is said to satisfy the *von Neumann inequality* if for every $p \in \mathbb{C}[z]$ and every $a \in \mathcal{A}$ such that $\|a\| \leq 1$, we have $\|p(a)\| \leq \sup\{|p(z)| : z \in \mathbb{D}\}$. Here $p(a)$ is defined by the usual polynomial functional calculus in a unital Banach algebra. It follows from the von Neumann inequality for $\mathcal{B}(H)$ that every Banach algebra which is an operator algebra, (i.e. which is isometrically isomorphic to a closed, not necessarily self-adjoint, subalgebra of $\mathcal{B}(H)$), also satisfies the von Neumann inequality. The converse question (is it true that every unital Banach algebra which satisfies the von Neumann inequality is an operator algebra ?) is an open question. A non-unital Banach algebra satisfying the von Neumann inequality for polynomials in a single variable, without constant term, and which is not (isomorphic to) an operator algebra can be found in [6].

Operator radii w_ρ . The class of operators $\mathcal{C}_\rho$ defined below has been introduced by Sz.-Nagy and Foiaş (see [18] Chapter 1] and the references therein). The operator $T \in \mathcal{B}(H)$ is said to be in the class $\mathcal{C}_\rho$ of operators admitting unitary ρ -dilations if there exists a larger space $\mathcal{H} \supset H$ and a unitary operator $U \in \mathcal{B}(\mathcal{H})$ such that

$$T^n = \rho P U^n P^*, \quad \text{for } n = 1, 2, \dots$$

The non-unital analogue is false, but the normalized unit is the obstruction to simply adjoining an identity to those counterexamples. The source's general unital question was still recorded as open in 2025 [4, p. 31].

2 Partial theorem

Proof intuition

Write an element as $\lambda 1 + x$, with $x^2 = 0$. Holomorphic functional calculus then remembers only the value and first derivative of a disk map: $f(\lambda 1 + x) = f(\lambda)1 + f'(\lambda)x$. A disk automorphism sending λ to zero expands the radical direction by exactly $(1 - |\lambda|^2)^{-1}$. Its inverse gives the reverse inclusion, so the entire unit ball is forced by its central slice. That forced ball is precisely the contraction ball of a 2×2 upper-triangular block operator. Embedding the Banach space X isometrically as diagonal Hilbert-space operators completes the representation.

Theorem 1 (square-zero local class). *Let $\mathcal{A}$ be a complex unital Banach algebra with $\|1\| = 1$. Suppose*

$$\mathcal{A} = \mathbb{C}1 \oplus X,$$

where X is a closed two-sided ideal and $X^2 = \{0\}$. If $\mathcal{A}$ satisfies von Neumann's inequality, then, writing $q(x) = \|x\|_{\mathcal{A}}$ for $x \in X$, one has

$$\|\lambda 1 + x\|_{\mathcal{A}} \leq 1 \iff |\lambda|^2 + q(x) \leq 1. \quad (1)$$

Consequently,

$$\|\lambda 1 + x\|_{\mathcal{A}} = \frac{q(x) + \sqrt{q(x)^2 + 4|\lambda|^2}}{2}. \quad (2)$$

Moreover, there are a Hilbert space H and a linear isometry $J : X \rightarrow \mathcal{B}(H)$ such that

$$\Pi(\lambda 1 + x) = \begin{pmatrix} \lambda I_H & Jx \\ 0 & \lambda I_H \end{pmatrix} \quad (3)$$

is a unital isometric algebra homomorphism $\mathcal{A} \rightarrow \mathcal{B}(H \oplus H)$. Its range is closed, so $\mathcal{A}$ is an operator algebra.

3 Proof

Lemma 2 (disk-algebra functional calculus). *If a is in the closed unit ball of a unital Banach algebra satisfying von Neumann's inequality, then*

$$\|f(a)\| \leq \|f\|_{\infty, \overline{\mathbb{D}}} \quad (f \in A(\overline{\mathbb{D}})).$$

In particular this holds for every automorphism of $\mathbb{D}$.

Proof. Choose polynomials converging uniformly to f on $\overline{\mathbb{D}}$. Applying the given inequality to differences shows that their values at a are Cauchy and that the limit has norm at most $\|f\|_{\infty}$. For a function holomorphic on a neighborhood of $\overline{\mathbb{D}}$, this limit agrees with the usual holomorphic functional calculus. $\square$

Lemma 3 (forced fibers of the unit ball). *Under the assumptions of Theorem 1, formula (1) holds.*

Proof. Let $a = \lambda 1 + x$ be in the closed unit ball. The quotient character $\mathcal{A} \rightarrow \mathcal{A}/X \cong \mathbb{C}$ has norm one, so $|\lambda| \leq 1$.

First suppose $|\lambda| < 1$ and set

$$\phi_\lambda(z) = \frac{z - \lambda}{1 - \overline{\lambda}z}.$$

Because $X^2 = 0$, polynomial calculus, followed by uniform approximation, gives

$$f(\lambda 1 + x) = f(\lambda)1 + f'(\lambda)x$$

for every function used here. Lemma 2 therefore yields

$$\|\phi_\lambda(a)\| = \left\| \frac{x}{1 - |\lambda|^2} \right\| \leq 1,$$

and hence $q(x) \leq 1 - |\lambda|^2$.

Conversely, assume $q(x) \leq 1 - |\lambda|^2$. Then $y = x/(1 - |\lambda|^2)$ is in the unit ball. The inverse disk automorphism

$$\psi_\lambda(z) = \frac{z + \lambda}{1 + \overline{\lambda}z}$$

satisfies $\psi_\lambda(0) = \lambda$ and $\psi'_\lambda(0) = 1 - |\lambda|^2$. Thus

$$\psi_\lambda(y) = \lambda 1 + x$$

is again in the unit ball by Lemma 2.

It remains to treat $|\lambda| = 1$. For each $n \geq 1$, von Neumann's inequality applied to z^n gives

$$\|\lambda^{-n} a^n\| = \|1 + n\lambda^{-1}x\| \leq 1.$$

Consequently $nq(x) \leq 2$, for every n , and $x = 0$. This proves both directions of (1). $\square$

The norm formula (2) now follows by taking the Minkowski functional of (1): the least $t > 0$ for which $|\lambda|^2/t^2 + q(x)/t \leq 1$ is the positive root of $t^2 - q(x)t - |\lambda|^2 = 0$.

Lemma 4 (upper-triangular contraction test). *For a Hilbert-space operator S and $\lambda \in \mathbb{C}$, put*

$$M_{\lambda,S} = \begin{pmatrix} \lambda I & S \\ 0 & \lambda I \end{pmatrix}.$$

Then

$$\|M_{\lambda,S}\| \leq 1 \iff |\lambda| \leq 1 \text{ and } \|S\| \leq 1 - |\lambda|^2.$$

Proof. For $|\lambda| < 1$, let $c = 1 - |\lambda|^2$. The upper-left corner of $I - M_{\lambda,S}^* M_{\lambda,S}$ is $cI > 0$, and its Schur complement is

$$cI - S^*S - \frac{|\lambda|^2}{c}S^*S = cI - \frac{1}{c}S^*S.$$

Thus positivity is equivalent to $S^*S \leq c^2I$. If $|\lambda| = 1$, positivity forces $S = 0$; if $|\lambda| > 1$, the spectral radius already exceeds one. $\square$

Proof of Theorem 1. Only the representation remains. Let K be the closed unit ball of X^* and put $H = \ell_2(K)$. The canonical diagonal map

$$(Jx)e_\varphi = \varphi(x)e_\varphi \quad (\varphi \in K)$$

is a linear isometry $X \rightarrow \mathcal{B}(H)$ by Hahn–Banach. Since $X^2 = 0$, direct block multiplication shows that (3) is a unital algebra homomorphism. Lemma 4, $\|Jx\| = q(x)$, and Lemma 3 show that Π preserves the closed unit ball exactly. Therefore Π is an isometry. Its range is closed because $\mathcal{A}$ is complete. $\square$

4 Verification, scope, and novelty audit

The block calculation was also checked in 2,000 seeded random complex matrix cases by `code/verify_block_contraction.py`; the largest discrepancy between the computed singular-value norm and formula (2) was at floating-point roundoff. This computation is only a regression check; the Schur-complement argument is the proof.

The theorem covers arbitrary-dimensional (including nonreflexive) square-zero radicals, but it requires a one-dimensional semisimple quotient and does not settle the source question for general radicals or multiple characters.

For novelty, the run searched its solution, attempt, proof-gap, and registry indexes for arXiv:1804.04062 and the core terms. Bounded exact-phrase searches for the unital question, two-dimensional algebras, dual numbers, square-zero ideals, and radical-square-zero algebras recovered Nakazi–Yamamoto’s complete two-dimensional theorem [2] and Paulsen–Solazzo’s semisimple $\mathbb{C}^k$ analysis [3], but no arbitrary-dimensional theorem with $\mathcal{A} = \mathbb{C}1 \oplus X$, $X^2 = 0$. Chirvasitu–Thompson still describe the general unital characterization as unknown in 2025 [4]. Novelty is therefore plausible but provisional; a specialist literature review is recommended.

Human-review focus. Verify the passage from polynomial to disk-algebra functional calculus, the two inclusions in the fiber formula, and whether the arbitrary-dimensional square-zero statement appears under different terminology in the operator algebra literature.

References

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